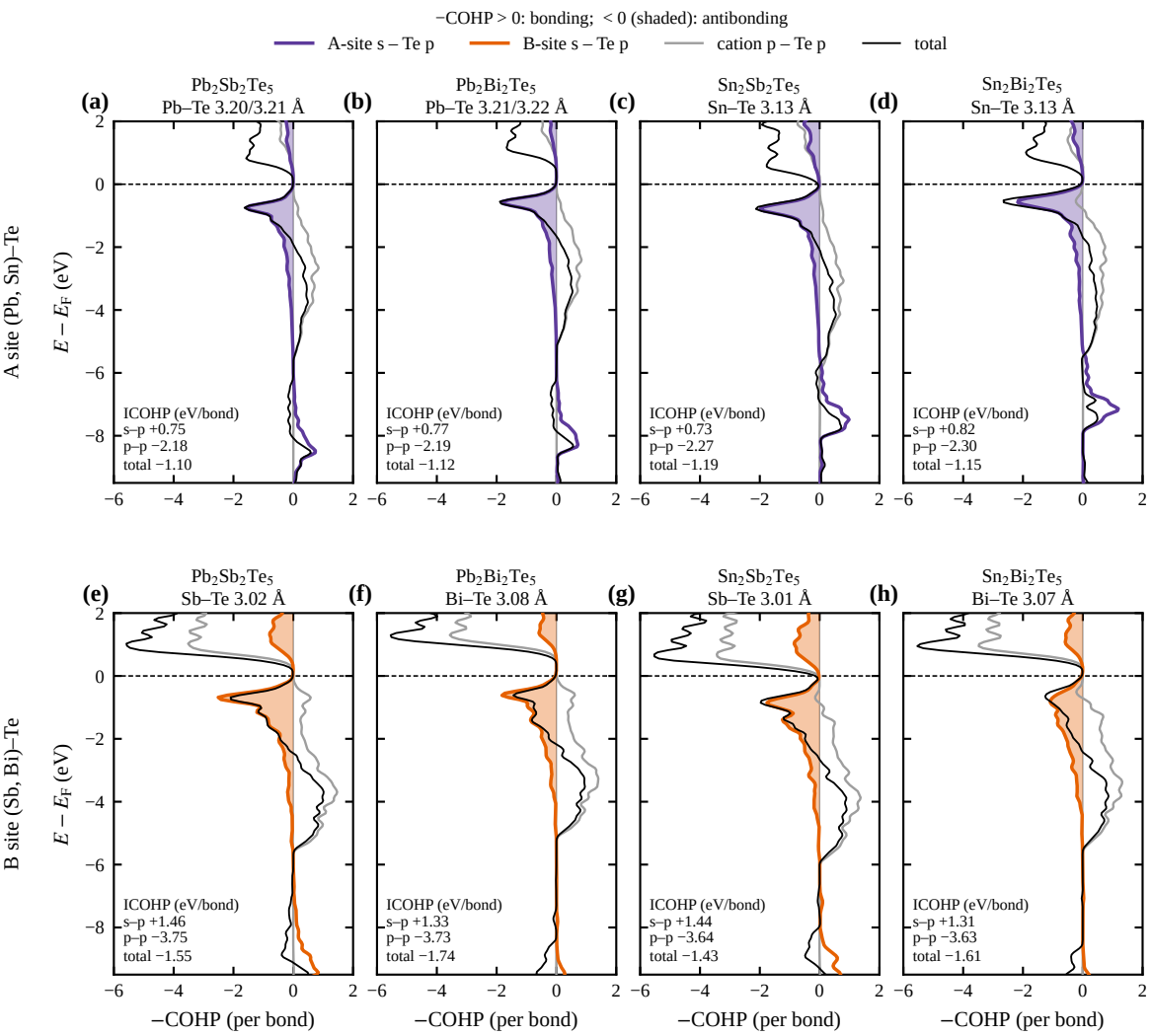

−COHP > 0: bonding; < 0 (shaded): antibonding
A-site s – Te p
B-site s – Te p
cation p – Te p
total
(a)
Pb2Sb2Te5
Pb–Te 3.20/3.21 Å
ICOHP (eV/bond)
s–p +0.75
p–p −2.18
total −1.10
(b)
Pb2Bi2Te5
Pb–Te 3.21/3.22 Å
ICOHP (eV/bond)
s–p +0.77
p–p −2.19
total −1.12
(c)
Sn2Sb2Te5
Sn–Te 3.13 Å
ICOHP (eV/bond)
s–p +0.73
p–p −2.27
total −1.19
(d)
Sn2Bi2Te5
Sn–Te 3.13 Å
ICOHP (eV/bond)
s–p +0.82
p–p −2.30
total −1.15
A site (Pb, Sn)–Te
E − EF (eV)
(e)
Pb2Sb2Te5
Sb–Te 3.02 Å
ICOHP (eV/bond)
s–p +1.46
p–p −3.75
total −1.55
(f)
Pb2Bi2Te5
Bi–Te 3.08 Å
ICOHP (eV/bond)
s–p +1.33
p–p −3.73
total −1.74
(g)
Sn2Sb2Te5
Sb–Te 3.01 Å
ICOHP (eV/bond)
s–p +1.44
p–p −3.64
total −1.43
(h)
Sn2Bi2Te5
Bi–Te 3.07 Å
ICOHP (eV/bond)
s–p +1.31
p–p −3.63
total −1.61
B site (Sb, Bi)–Te
E − EF (eV)
−COHP (per bond)
−COHP (per bond)
−COHP (per bond)
−COHP (per bond)